

[1] Overview

Microcontroller

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Sheet: ESP32

3

Power

Sheet: Power

4

4

Audio

Sheet: Audio

5

IMU

Sheet: IMU

6

I/O

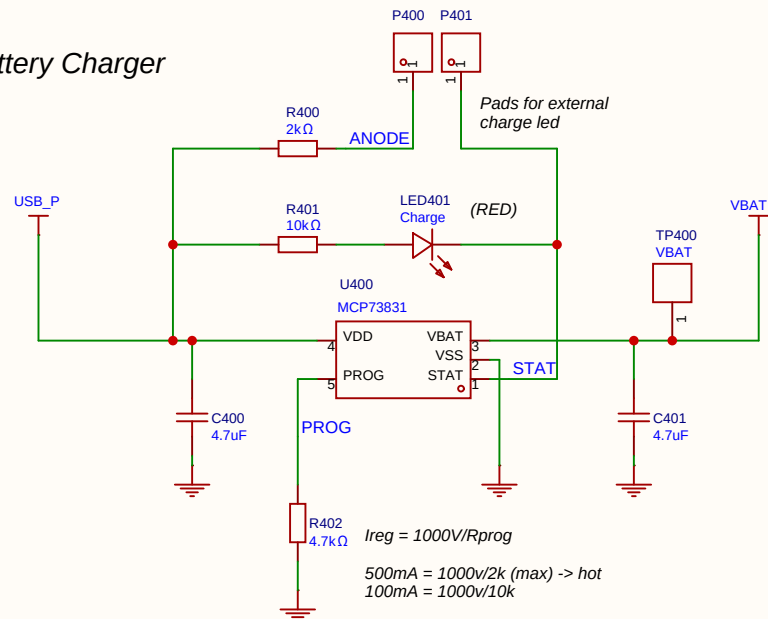
Sheet: Connectors / IO

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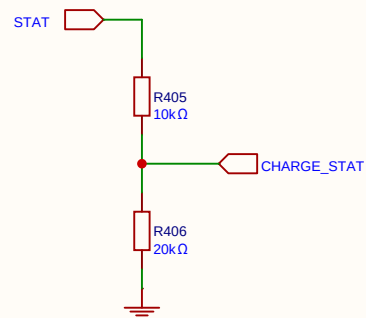
TITLE: RocketShooter_from_STD		REV: 1.0
EasyEDA	Company: Hobbyist	Sheet: 1/7
	Date: 2023-06-25	Drawn By: christoph.streit

[4] Power

Battery Charger

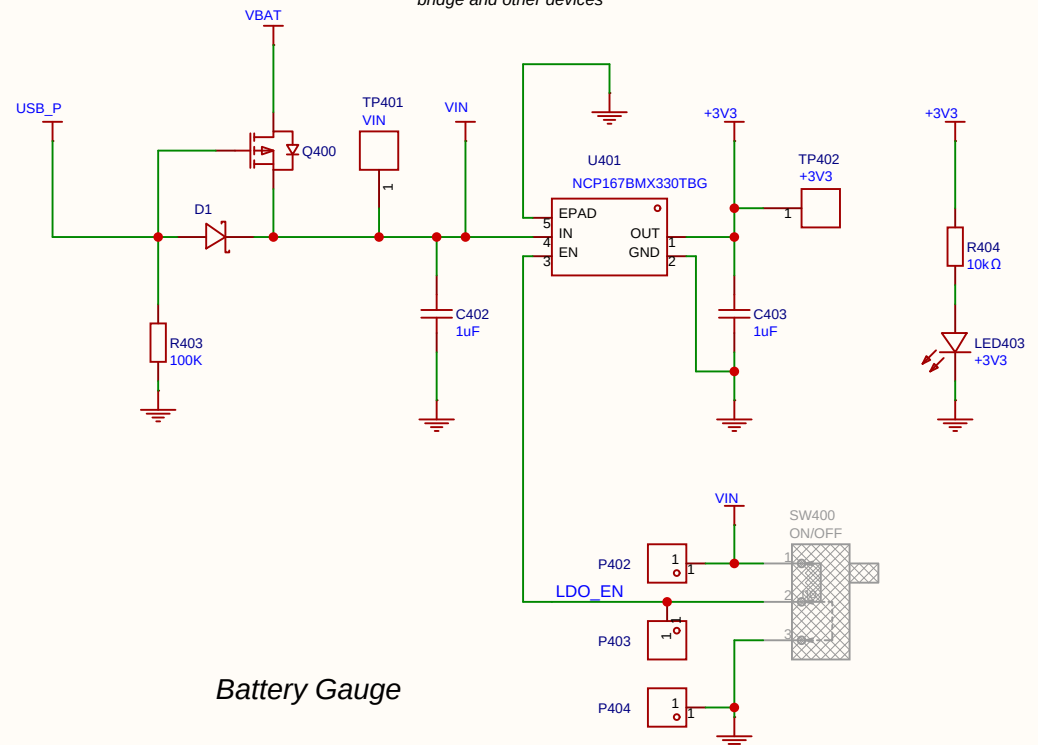


Charge Sense

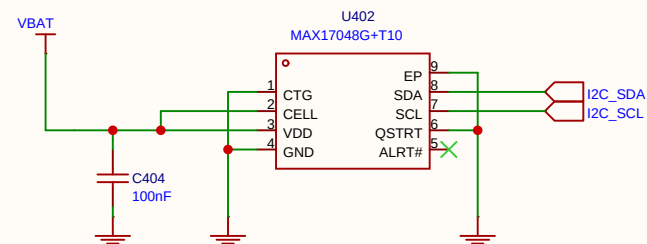


3.3V LDO

3.3V LDO to power the ESP, the CP2102 bridge and other devices



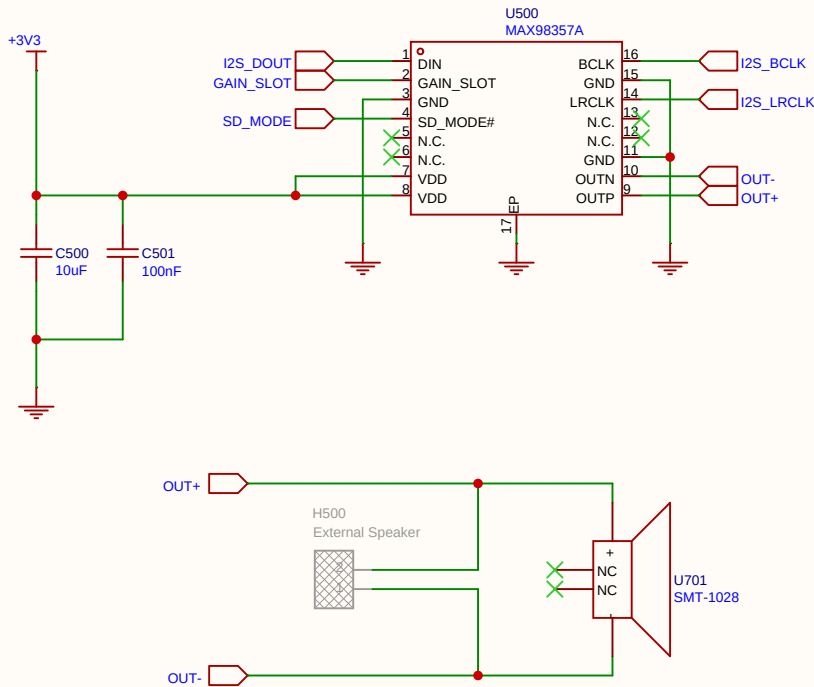
Battery Gauge



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EasyEDA		Company: Hobbyist
Date: 2023-11-03		Sheet: 4/7
Drawn By: christoph.streit		

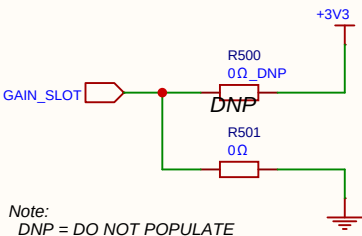
[5] Audio

Amplifier



Gain Selection

GAIN_SLOT	GAIN
Connect to GND through 100kΩ resistor	+15 dB
Connect to GND	+12 dB
Unconnected	+9 dB
Connect to VDD	+6 dB
Connect to VDD through 100kΩ resistor	+3 dB

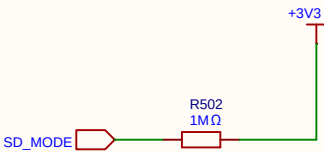


Channel Selection (SD_MODE)

LEFT = VDD (0 Ω)

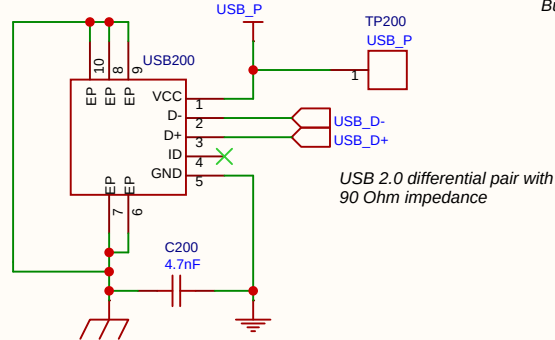
RIGHT = 370K
= $94 \times 5 - 100$ (RSMALL (kΩ) = $94.0 \times VDDIO - 100$)

LEFT/2 + RIGHT/2 = 1.011M
= $222.2 \times 5 - 100$ (RLARGE (kΩ) = $222.2 \times VDDIO - 100$)

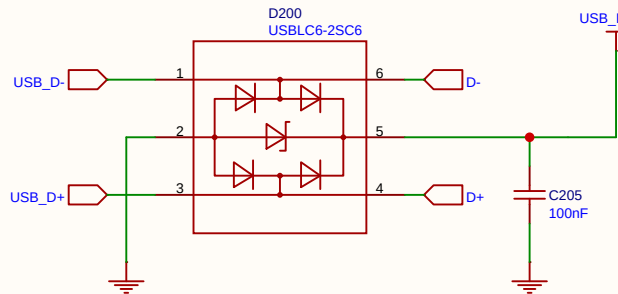


[2] USB & Serial Bridge

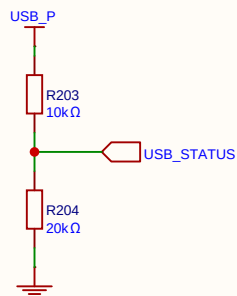
USB



ESD Protection

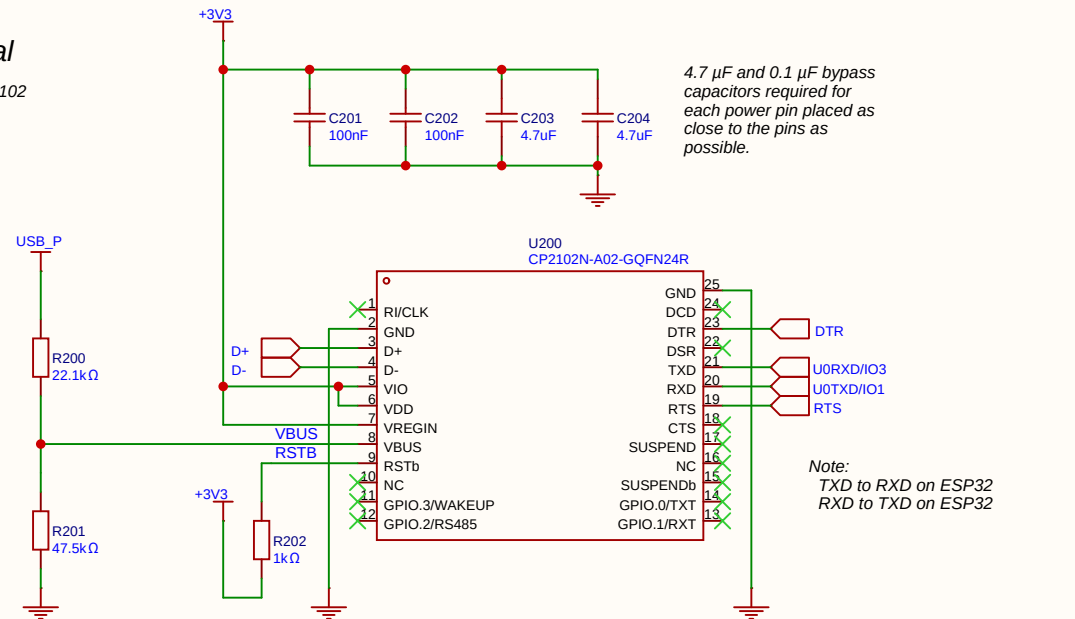


USB Sense



USB to Serial

Bus-powered configuration of the CP2102

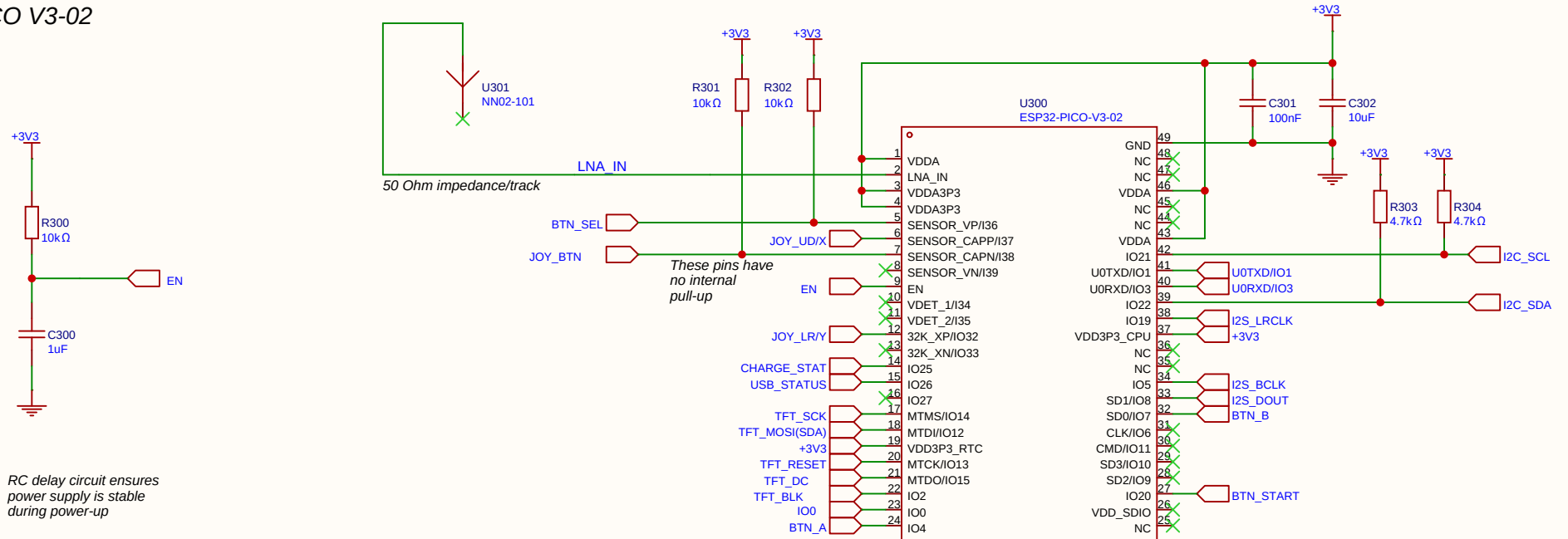


Note:
TXD to RXD on ESP32
RXD to TXD on ESP32

Both SUSPEND and SUSPENDb temporarily float high during a CP2102N reset. If this behavior is undesirable, a strong pull-down (10 k Ω) can be used to ensure SUSPENDb remains low during reset.
--> not installed

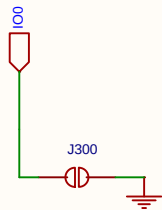
[3] ESP32

ESP32 PICO V3-02

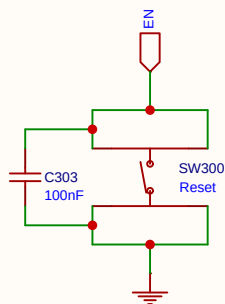


Datasheet: https://www.espressif.com/sites/default/files/documentation/esp32-pico-v3-02_datasheet_en.pdf

Boot Option and Reset Button

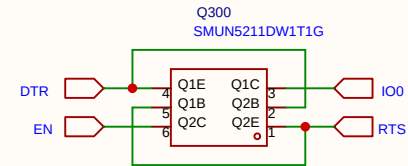


Boot Option to put the board in flashing mode, e.g. short the solder jumper with tweezers.



Press reset button to reset (restart) the board.

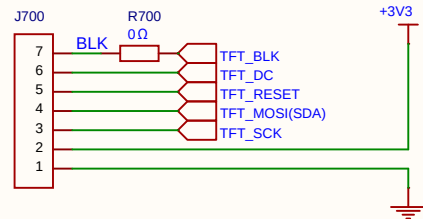
Autoreset



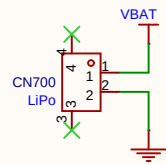
TITLE: RocketShooter_from_STD		REV: 1.0
Company: Hobbyist		Sheet: 3/7
Date: 2023-10-21	Drawn By: christoph.streit	

[7] Connectors / IO

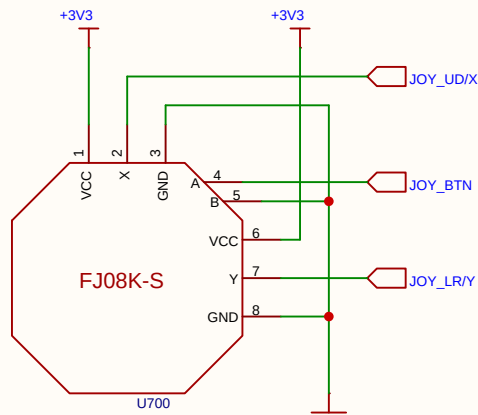
Display



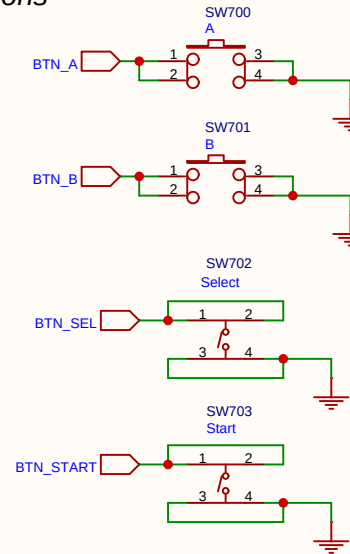
LiPo



Joystick

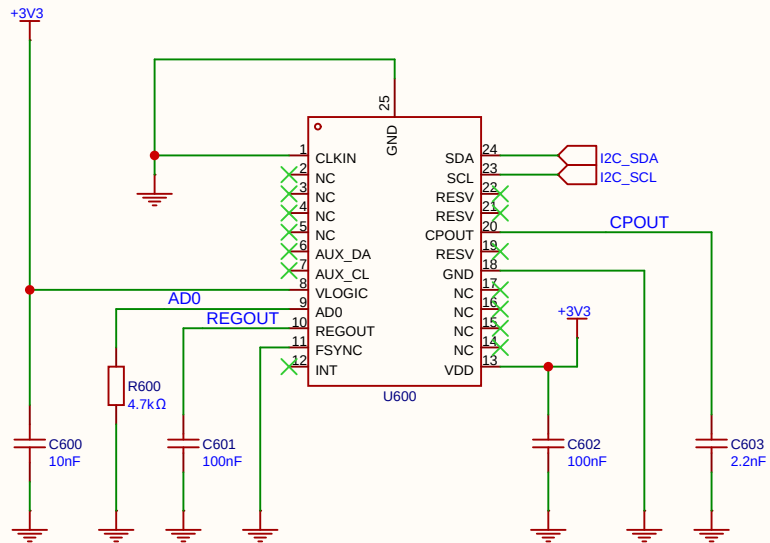


User Buttons



TITLE: RocketShooter_from_STD		REV: 1.0
Company: Hobbyist		Sheet: 7/7
Date: 2023-10-21	Drawn By: christoph.streit	

[6] IMU



TITLE: RocketShooter_from_STD		REV: 1.1
EasyEDA	Company: Hobbyist	Sheet: 6/7
	Date: 2023-11-19	Drawn By: christoph.streit